

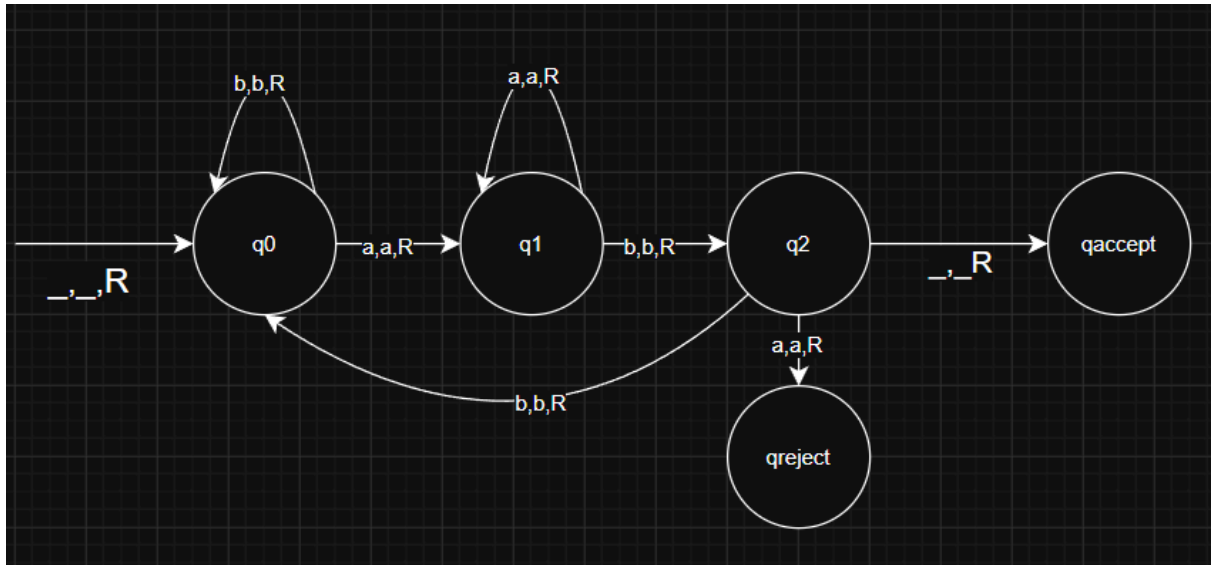
Workshop 3 .1 Turing Machine and decidable

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1) Given the language:

$$L = \{w \mid w \in \Sigma^*, \Sigma = \{a, b\}, w \text{ does not contain the substring } aba\}$$

a) Construct a Turing Machine that accepts L.



b) Trace the execution for the input bababb

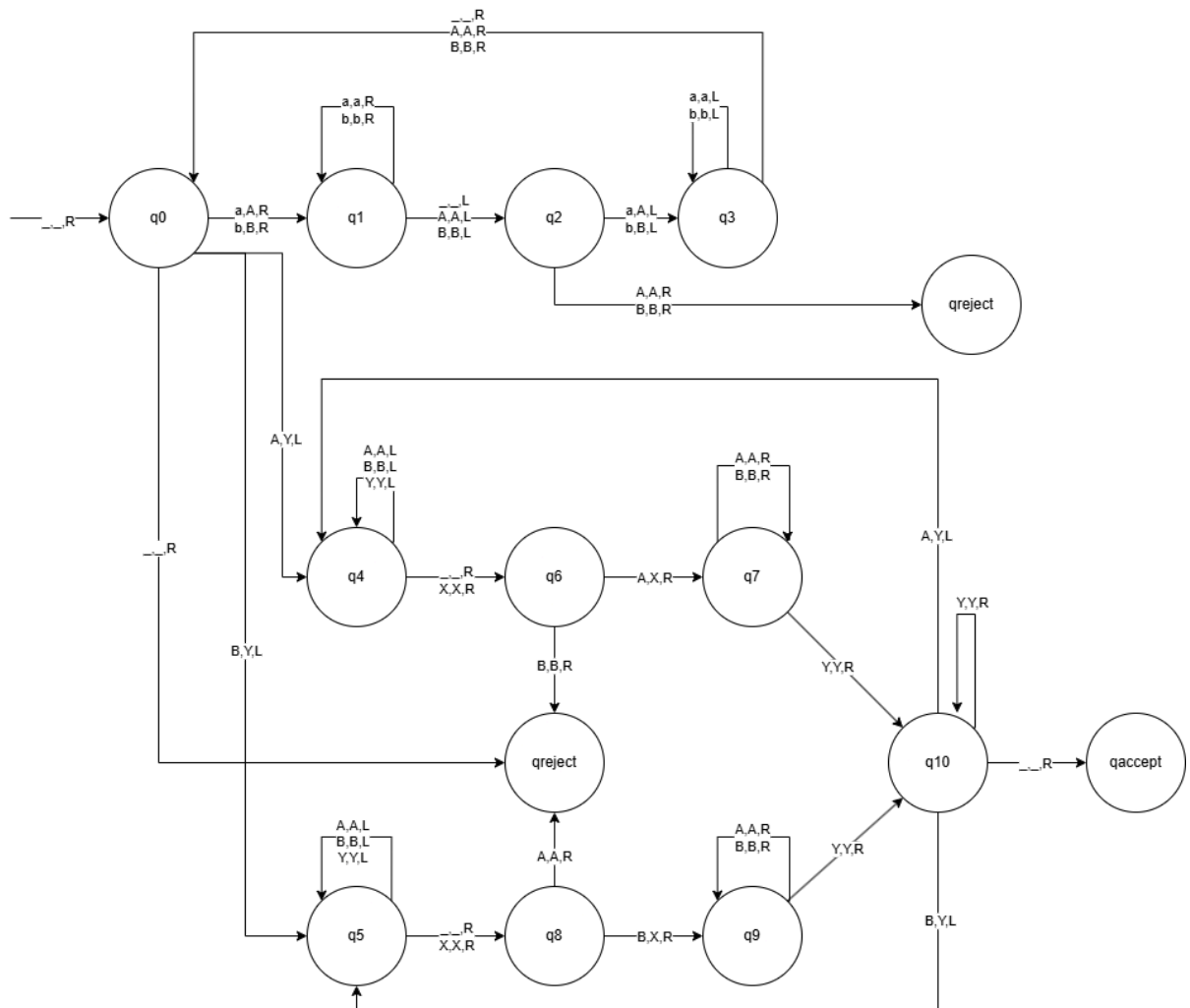
state	input - cursor (bold one)	decisión
q0	b ababb	
q0	a b abb	
q1	ba b b	
q2	ab b	
qreject	bb	the substring aba is in the string REJECTED

c) Show that L is decidable

- yes, because it only goes right til the input end or the substring aba is in the string.

2) Given the language $L = \{ww \mid w \in \{a,b\}^+\}$

a) Design a Turing machine that accepts L



b) Describe how the designed TM works

- first we find the first character of the second "String" then we prove that the string is the same

c) Trace the execution for the input abab

State	Input - Cursor (Bold one)	Decisión
q0	_abab_	
q1	_Abab_ _Abab_ _Abab_ _Abab_	

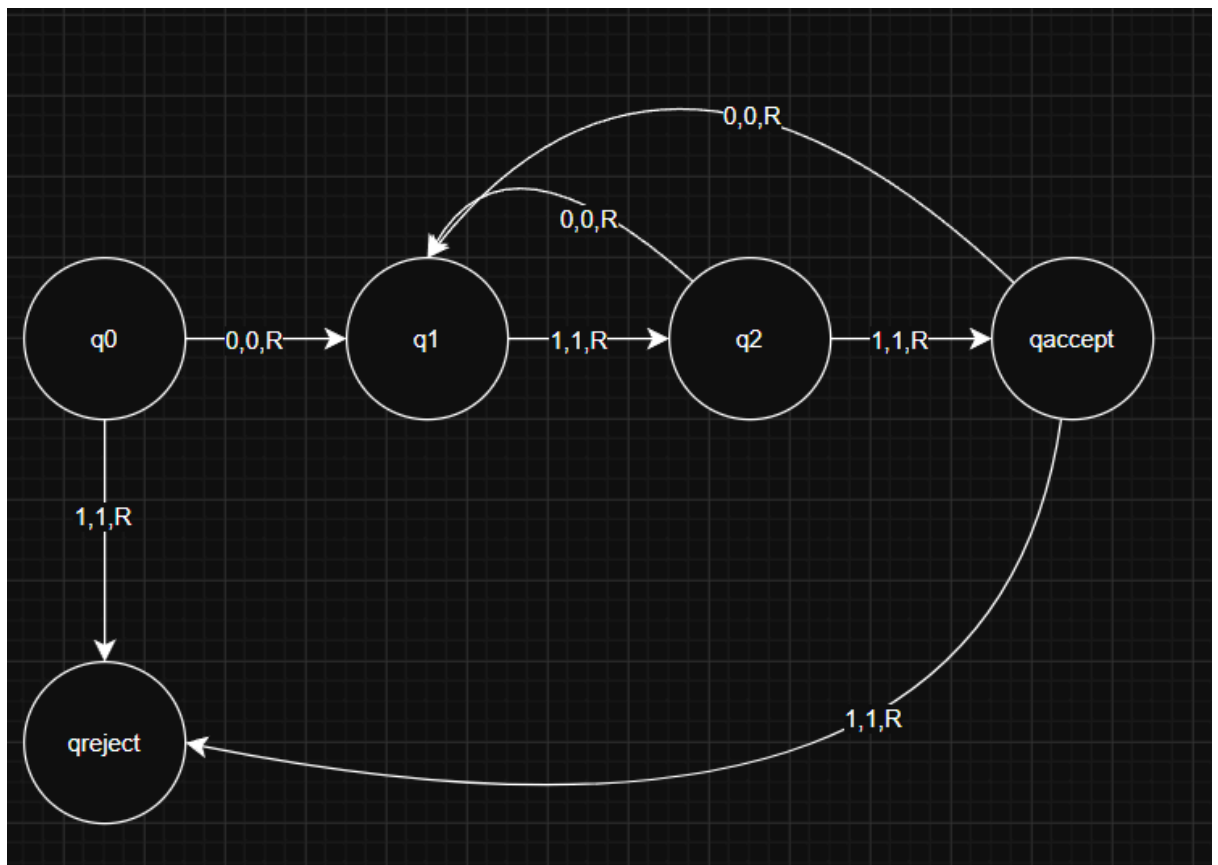
q2	_Abab_	
q3	_AbaB_ _AbaB_ _AbaB_	
q0	_AbaB_	
q1	_ABaB_ _ABaB_	
q2	_ABaB_	
q3	_ABAB_	
q0	_ABAB_	
q4	_ABYB_ _ABYB_ _ABYB_	
q6	_ABYB_	
q7	_XBYB_ _XBYB_	
q10	_XBYB_	
q5	_XBYY_ _XBYY_ _XBYY_	
q8	_XBYY_	
q9	_XXYY_	
q10	_XXYY_ _XXYY_	
qaccept	_XXYY_	The string is accepted

d) Trace the execution for the input bab

State	Input - Cursor (Bold one)	Decisión
q0	_bab_	
q1	_Bab_ _Bab_ _Bab_	
q2	_Bab_	

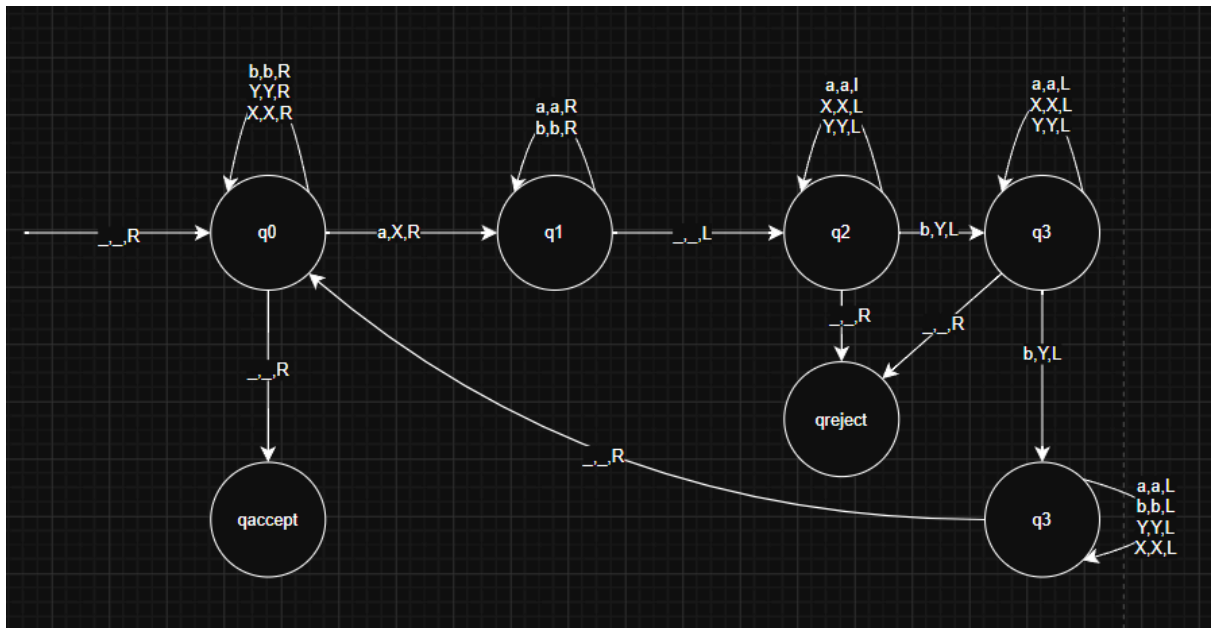
q3	<u>BaB</u> <u>BaB</u>	
q0	<u>BaB</u>	
q1	<u>BAB</u>	
q2	<u>BAB</u>	
qreject	<u>BAB</u>	rejected because the length of the string is odd.

3) Design a turing machine with no more than 5 states that accepts the language $(01 + 011)^*$



4) Given the language $L = \{ w \in \{a, b\}^* \mid \#a(w) \leq 2\#b(w) \}$

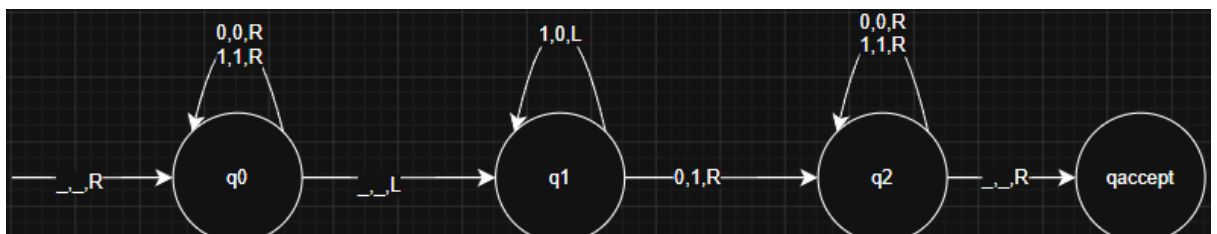
a) Design a turing machine that accepts L



b) Is L decidable? justify your answer

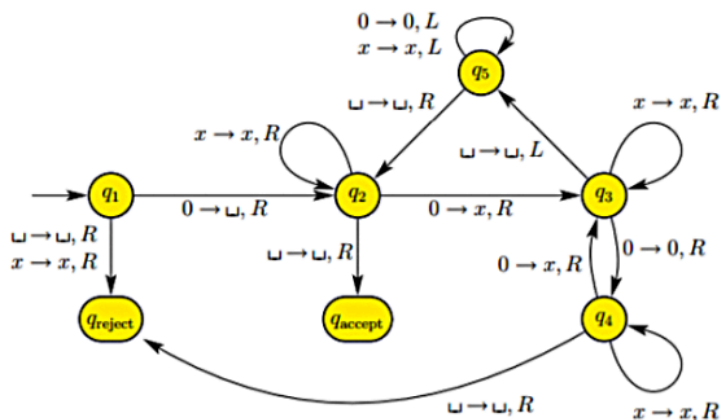
yes, because always have an answer for this, or the string ends, or encounters an extra a so always is going to stop.

- 5) Design a turing machine that adds one to a given binary number. The input binary number is written on the tape, surrounded by blank symbols. The machine must halt in an accept state with the incremented binary number left on the tape.



- 6) The turing machine showed in the next figure recognizes the language

$$L = \{0^{2^n} \mid n \geq 0\}$$



a) Analyze how the machine processes the strings:

- 0000000

state	input - cursor (bold one)	decisión
q1	0 000000	
q2	_000000	
q3	_x 0 0000	
q4	_x0 0 000	
q3	_x0x 0 00	
q4	_x0x0 0	
q3	_x0x0x 0	
q4	_x0x0x0 _	
qreject	_x0x0x0__	rejected

- 000000

State	Input - cursor (bold one)	Decisión
q1	0 00000	
q2	_00000	
q3	_x 0 000	
q4	_x0 0 00	
q3	_x0x 0 0	
q4	_x0x0 0	
q3	_x0x0x _	
q5	_x0x0x _x0x 0 x _x0x0x _x 0 x0x _x0x0x _x0x0x	

q2	_x0x0x _x0x0x	
q3	_xxx0x _xxx0x	
q4	_xxx0x _xxx0x_	
qreject	_xxx0x__	rejected

- 00000

State	Input - Cursor (bold one)	Decisión
q1	00000	
q2	_0000	
q3	_x000	
q4	_x000	
q3	_x0x0	
q4	_x0x0_	
qreject	_x0x0__	

7) Given the language $L = \{ w\#x \mid w \text{ is a substring of } x \}$

I don't know the alphabet.

- design a turing machine that accepts L
- Describe how the designed TM woks

8) Given the language $L = \{ x + y + z \mid x, y, z \text{ are binary integers and } x + y = z \}$

- Design a Turing machine that accepts L
- Describe how the designed TM works